LAB 4: Activity Diagram – Library Management System

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Aim

To draw the activity diagram of the Library Management System showing the flow of activities for borrowing and returning a book.

Theory

An activity diagram is a behavioural UML diagram that shows the flow of control from one activity to another. It is similar to a flowchart but it can also show parallel work and the person or system responsible for each activity (using swimlanes).

The elements used in the diagram are:

- **Initial node:** a filled circle that shows where the process starts.
- **Activity (action):** a rounded rectangle that shows a task performed in the process.
- **Decision node:** a diamond with one incoming flow and two or more outgoing flows, each marked with a condition such as Yes or No.
- **Control flow:** an arrow that connects one node to the next.
- **Swimlane (partition):** a vertical lane that shows who performs the activities placed inside it.
- **Final node:** a circle with a filled centre that shows where the process ends.

Process Description

The process drawn is borrowing and returning a book in the Library Management System. It involves three swimlanes: Member, Librarian and System.

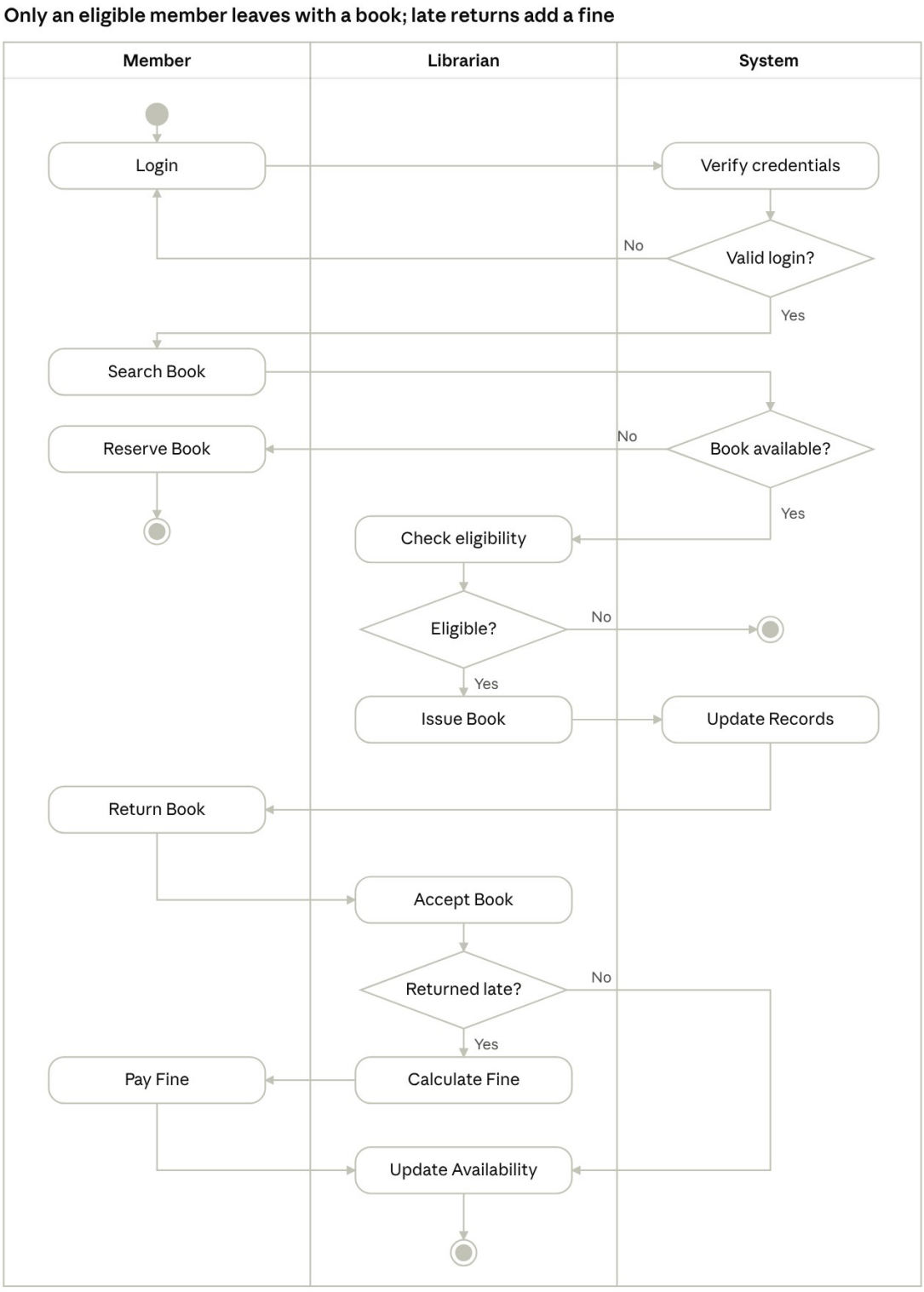
- The member logs in and the system verifies the credentials. If the login is invalid, the member has to log in again.
- The member searches for a book. If the book is not available, the member reserves it and the process ends.
- If the book is available, the librarian checks whether the member is eligible (no pending fine and the issue limit is not crossed). If the member is not eligible, the request is rejected and the process ends.
- If the member is eligible, the librarian issues the book and the system updates the records.
- When the member returns the book, the librarian accepts it and checks whether it is late. For a late return the fine is calculated and the member pays it.
- Finally the librarian updates the availability of the book and the process ends.

Activities and Responsible Actors

Step	Activity	Performed by
1	Login	Member
2	Verify credentials (decision: valid login?)	System
3	Search Book	Member
4	Check availability (decision: book available?)	System
5	Reserve Book (when the book is not available)	Member
6	Check eligibility (decision: eligible?)	Librarian
7	Issue Book	Librarian
8	Update Records	System
9	Return Book	Member
10	Accept Book (decision: returned late?)	Librarian
11	Calculate Fine (for a late return)	Librarian
12	Pay Fine	Member
13	Update Availability	Librarian

Activity Diagram

The activity diagram of borrowing and returning a book is shown below.



activity diagram · 3 swimlanes, 4 decisions

Figure 1: Activity diagram for borrowing and returning a book in the Library Management System.

Conclusion

The activity diagram of the Library Management System was drawn using swimlanes for the Member, Librarian and System. It shows the complete flow from login to return of the book, including the decisions for valid login, book availability, member eligibility and late return.